MAUVE-MUSE: When Metallicity Follows or Fights Star Formation

A Mass Dependent Inversion in Virgo Spirals

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INTRODUCTION

Resolved Mass-Metallicity Relation (rMZR) probes the balance of **metal enrichment vs dilution** in discs.

Key question: Does gas-phase metallicity (**Z_{gas}**) **correlate or anti-correlate with** Star formation rate (SFR) surface density (**Σ_{SFR}**) at fixed stellar mass surface density (Σ_*), and does the sign changes across discs?

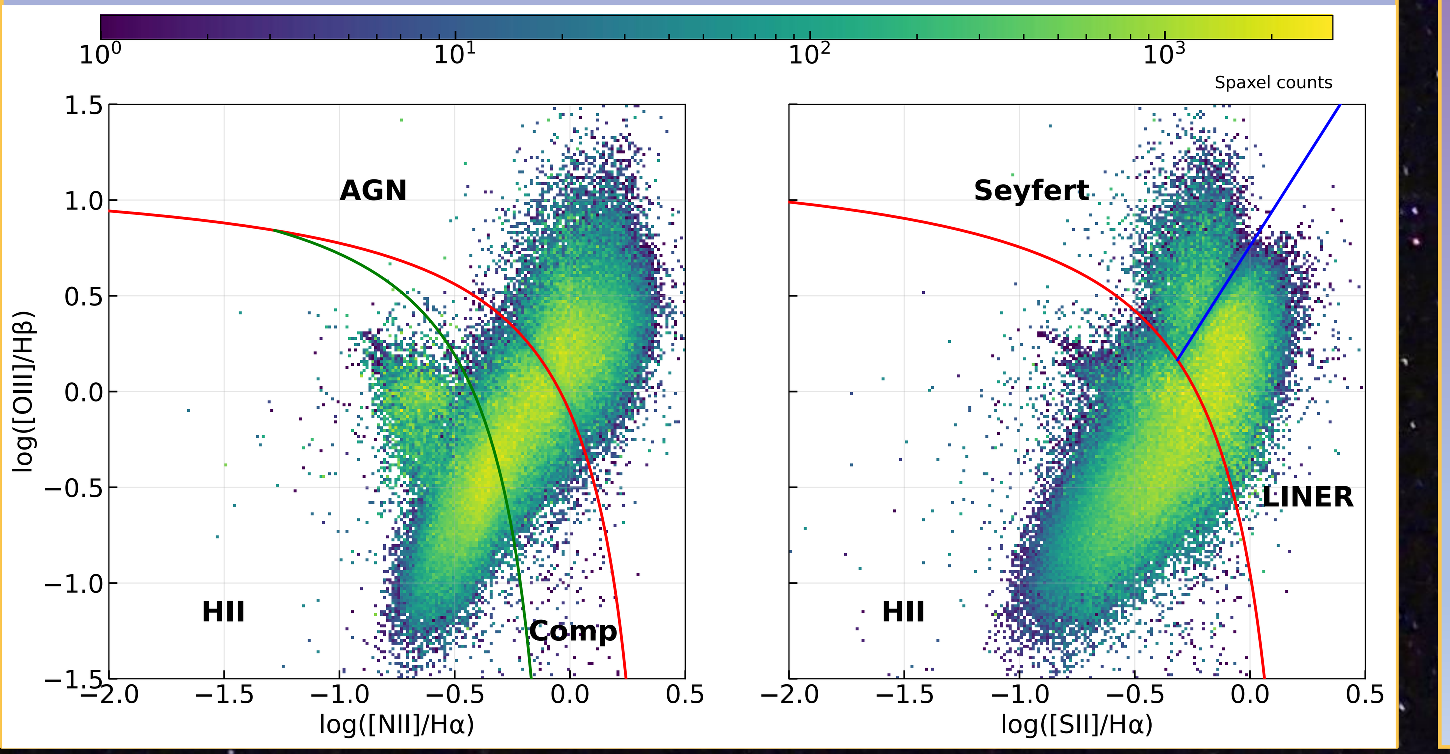
DATA & METHODS

Data: MAUVE–MUSE (Virgo, 100 pc)

1. **Instrument:** VLT/MUSE wide-field IFU, delivering stellar mass and emission-line maps on ~100 pc scales in Virgo spirals;
2. **Sample: 14 Virgo spirals** (MAUVE–MUSE **v2 internal release**); survey observations are **ongoing;**
3. **Working set:** fiducial ensemble uses **13 galaxies** (excluding the metallicity outlier **NGC 4383**);
4. **Resolved Maps used (co-spatial spaxels): Σ_*, Σ_{SFR}** and **Z_{gas}**.

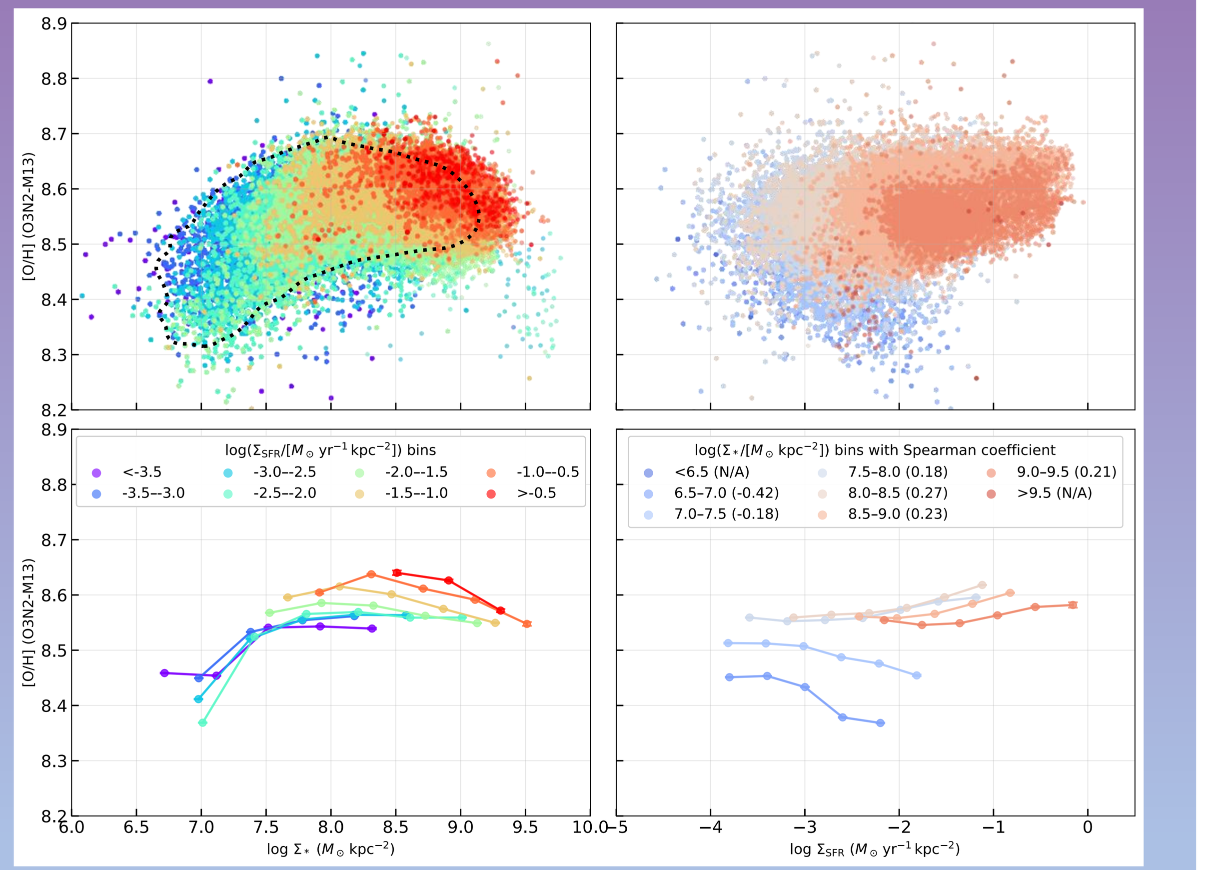
Method: isolate star-forming spaxels and measure **Z_{gas}**

1. **Star-Forming selection:** keep **H II** spaxels using [N II] and [S II] **BPT selection** diagnostic cuts;
2. **Calibration:** fiducial **Z_{gas}** from **O3N2–M13** (Marino+2013) using strong lines (dust-corrected Hα Hβ, [O III], [N II]);
3. **ΔZ_{gas}- $\Delta\Sigma_{SFR}$** coupling measurement: remove primary mass-dependent trends, then measure residual coupling:
   $\rho(\Sigma_*) = \mathrm{corr}(\Delta Z_{gas}, \Delta\log\Sigma_{SFR})|_{\Sigma_*}$.



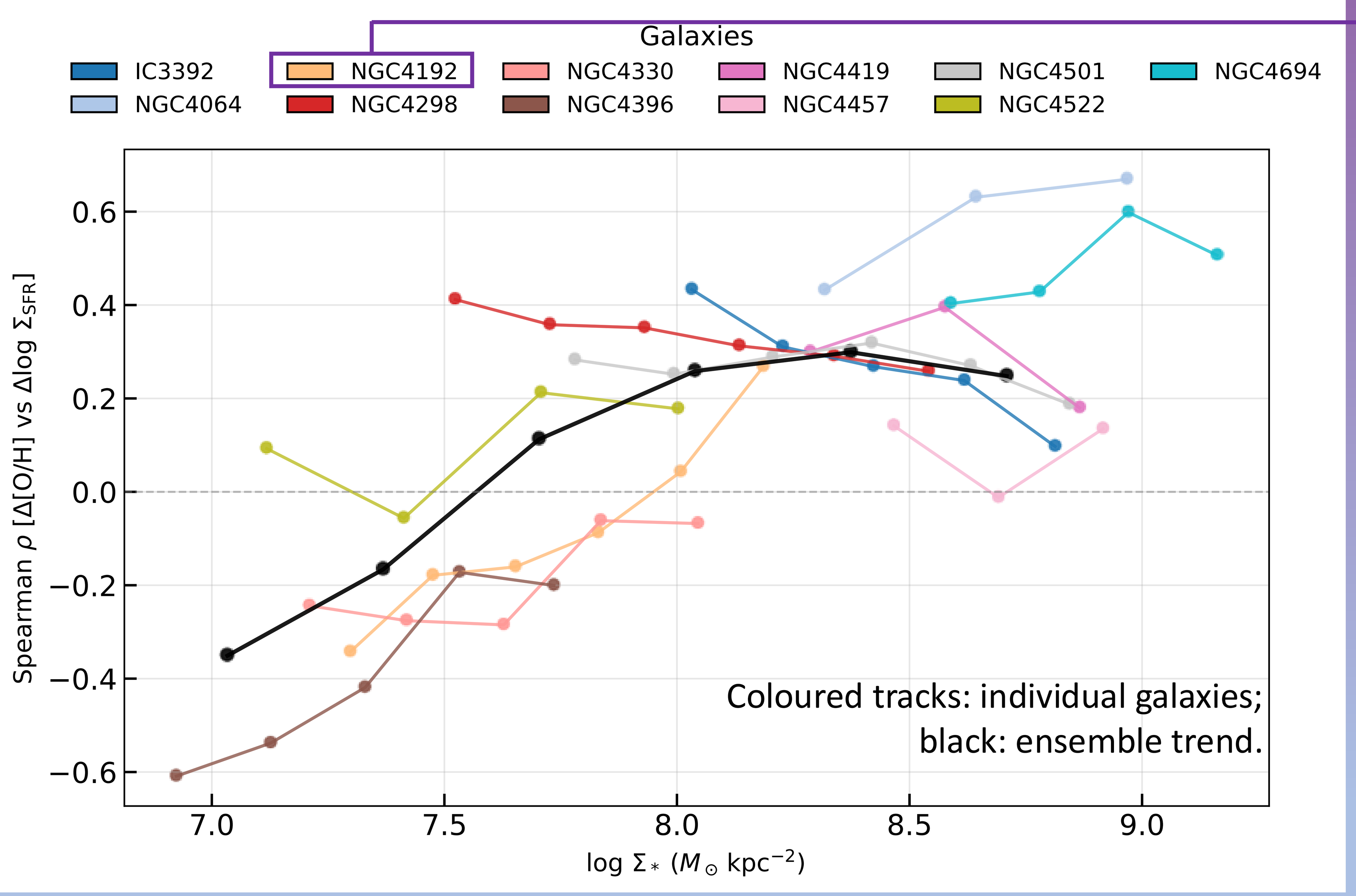
RESULTS

1. rMZR is well-defined;
2. The secondary **ΔZ_{gas}- $\Delta\Sigma_{SFR}$** relation depends on **Σ_***.

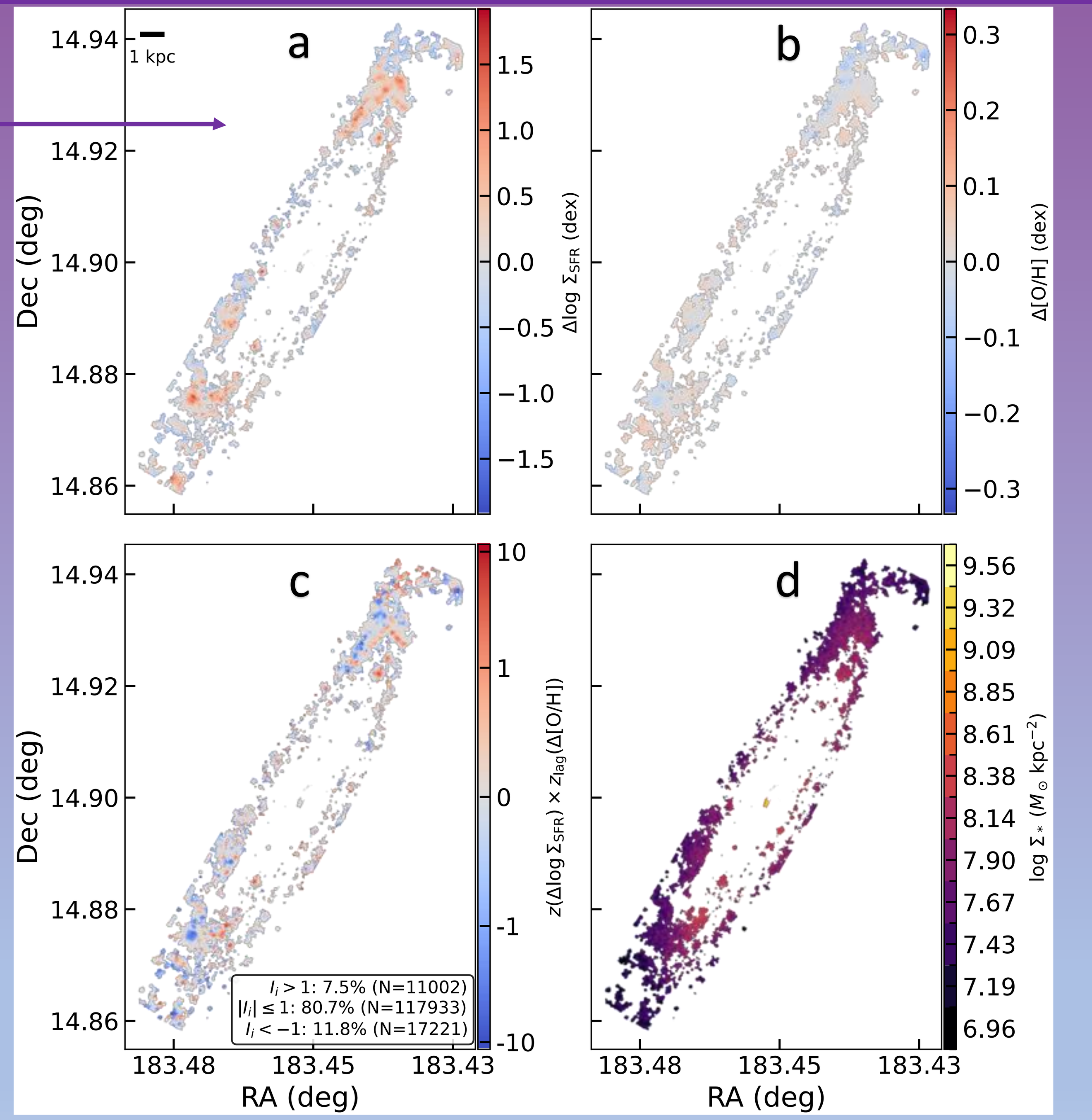


Left: rMZR: **Z_{gas}** rises with **Σ_***; colour shows **Σ_{SFR}**.
Right: within narrow **Σ_*** bins, the **Z_{gas} − Σ_{SFR}** trend changes sign: **low Σ_* anti-correlation → high Σ_* positive correlation**.

Example (**NGC 4192**): correlated and anti-correlated regions coexist locally



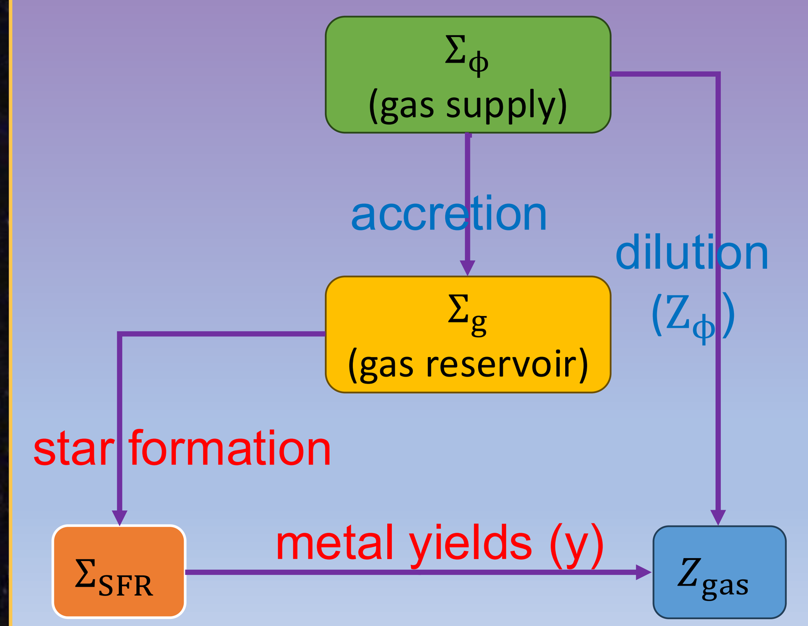
We compute a Spearman rank correlation between residual ΔZ_{gas}- $\Delta\Sigma_{SFR}$. Across galaxies, Σ_* transitions smoothly from **negative** (gas-supply-dominated) to **positive** (star-formation-dominated), with an inversion point at $\log(\Sigma_*/M_\odot \mathrm{kpc}^{-2}) \approx 7.5 - 8.0$



Panels a,b,d: **Σ_{SFR}, Z_{gas}** and **Σ_*** maps. Panel c: A local Moran-like index quantifies neighbourhood co-variation between **ΔZ_{gas}** and **$\Delta\Sigma_{SFR}$**. **red = concordant patches**, **blue = discordant patches.**

INTERPRETATION: LOCAL GAS-REGULATOR

$$\frac{d}{dt}\Sigma_g = \Sigma_\phi - (1-R)\Sigma_{SFR} - \Sigma_{out};\ \frac{d}{dt}Z_{gas} = \frac{1}{\Sigma_g}[y(1-R)\Sigma_{SFR} + (Z_\phi - Z_{gas})\Sigma_\phi]$$



A spatially resolved *bathtub* model to describe the local gas regulation.

At fixed Σ_*:

$$\Delta Z_{gas} \propto \Delta\log\Sigma_{SFR} - \Delta\log\Sigma_\phi$$

Star-formation-driven variability dominates $\Rightarrow \uparrow \Sigma_{SFR}$ (metal yields) with smaller changes in $\Sigma_\phi \Rightarrow \uparrow Z_{gas}$.
Gas-supply-driven variability dominates $\Rightarrow \uparrow \Sigma_{SFR}$ with with larger changes in Σ_ϕ (Z_ϕ dilution) $\Rightarrow \downarrow Z_{gas}$.

CONCLUSIONS

1. **Detect** a mass-dependent inversion in residual **ΔZ_{gas}- $\Delta\Sigma_{SFR}$** coupling;
2. **Show** that this flip-sign is spatially varying in the local Moran-like map;
3. **Interpret** as a transition from **gas supply/dilution-dominated** to **star formation/enrichment-dominated** regulation;
4. **Implication**: this resolved trends unify with integrated metallicity–SFR behaviour.

ROBUSTNESS

1. The **sign flip persists** across O3N2-based calibrations;
2. HII+Comp BPT selection **keeps the same Σ_* transition**;
3. Metallicity outlier, NGC4383, points out an **opposite behavior** in outflows/extraplanar components and/or strong, recent gas-supply perturbations.

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